# Emergency Response Plan

## Executive Statement

Safety, environmental protection, and legal compliance take precedence over production. This Plan is compulsory. Any deviation without prior written approval from the NPA Safety Director and the GK & A Managing Director is a breach subject to disciplinary action, removal from site, and contract termination for cause.

## Introduction

This Plan is a jointly endorsed, auditable operational control developed under Nigerian Ports Authority (NPA) oversight. Procedures, notification chains, and role interfaces have been reviewed with NPA and are mandatory for all personnel and contractors on GK & A-controlled areas of the NPA Lighter Terminal, Ikorodu.

| Aspect | Description |
|---|---|
| Enforcement | Roles and accountabilities are assigned in writing. Compliance is continuously monitored. Violations are escalated through disciplinary protocols and may trigger regulatory reporting to NPA and relevant authorities. |
| Joint Execution | GK & A and NPA Safety Officers shall co-lead emergency activations, joint drills, audits, debriefs, and corrective actions. NPA retains authority to intervene and direct operations where necessary to protect life, environment, and port security. |

## 1. Purpose

This Plan establishes an integrated, zero-harm emergency management framework to:

| Objective | Description |
|---|---|
| Protection | Protect people, property, cargo integrity, and the environment |
| Prevention | Prevent escalation and minimize business interruption |
| Compliance | Achieve and evidence compliance with NPA directives, ISPS Code, IMDG Code, ISO 45001 and ISO 14001 principles, ILO/IMO Code of Practice for Safety and Health in Ports, Lagos State and Federal environmental obligations |
| Standards | Apply the most stringent requirement when standards overlap |

### Measurable Objectives

| Metric | Target |
|---|---|
| Fatalities | Zero |
| Serious Injuries | Zero (life-altering) |
| Notification Time | Initiated within 5 minutes of emergency declaration |
| Headcount Reconciliation | Full headcount within 10 minutes of evacuation order |
| Drill Participation | ≥ 90% of on-duty personnel |
| Corrective Action Closure | ≤ 30 days |
| Spill Containment | No uncontrolled discharge to water; all spills contained within site boundary |

## 2. Scope

Applies 24/7 to GK & A-controlled operations within the NPA Lighter Terminal, Ikorodu:

| Area | Description |
|---|---|
| Primary Operations | Quay, jetties, cargo yards, and operational zones under NPA regulation |
| Storage | Warehouses and open storage areas |
| Refrigeration | Reefer container stacks, plug-in racks, and associated power/refrigeration systems |

### Interfaces and Inclusions

| Group | Coverage |
|---|---|
| Personnel | All employees, contractors, subcontractors, visitors, clients, and third-party carriers |
| Third Parties | Tenants or third parties while in GK & A-controlled zones |
| Regulatory Oversight | NPA Safety Officers participate in activations, drills, debriefs, and reviews |

### Exclusions

| Category | Details |
|---|---|
| Non-GK & A Premises | Excludes non-GK & A premises beyond defined interface points |
| Mutual Aid | GK & A shall support NPA-directed mutual aid and information sharing as requested |

## 3. Definitions

| Term | Definition |
|---|---|
| Emergency | Unplanned event threatening life, environment, assets, cargo, or operations |
| Incident Commander (IC) | On-scene authority responsible for declaring and managing the incident; approves re-entry |
| Deputy IC | Authorized alternate who assumes command if needed |
| Emergency Response Team (ERT) | Trained responders for Fire, HAZMAT/Spill, Medical, Refrigerant/Technical, Security |
| Public Information Officer (PIO) | Sole authorized spokesperson for external communications |
| Muster Point | Designated safe assembly location for headcount after evacuation |
| Exclusion Zone | Controlled perimeter restricting access to trained, authorized personnel only |
| ALARP | As Low As Reasonably Practicable; risk threshold for operations and rescue |
| Refrigerant | Substances used in reefers (e.g., A1: R-134a, R-404A, R-452A; A2L where applicable) |
| Shelter-in-place | Controlled indoor protection procedure during external airborne or security threats |
| Command Post (CP) | Primary/secondary locations where IC and key staff coordinate response |
| Staging Area | Controlled location where resources and teams assemble before tasking |

## 4. Regulatory and Standards Alignment

This Plan aligns with and references:

| Regulation/Standard | Description |
|---|---|
| NPA Safety Directives | Safety directives, permits, and port facility requirements |
| ISPS Code | Port Facility Security Plan (PFSP) interfaces |
| IMDG Code | SDS obligations; dangerous goods segregation |
| Environmental Laws | Lagos State and Federal environmental laws; LASEPA/NESREA notifications |
| ILO/IMO Code | Code of Practice for Safety and Health in Ports |
| LASEMA/NEMA | Incident coordination guidance |
| LSSC | Lagos State Safety Commission guidance |
| LASWA/NIWA | Coordination for waterway/marine incidents |
| Good Practice | NFPA firefighting guidance; EN 378 for refrigerating systems and machinery rooms (where applicable) |
| Data Protection | Nigeria Data Protection Act (NDPA) for incident/CCTV data handling |

Where requirements differ, the stricter requirement applies.

## 5. Risk Profile and Major Accident Hazards (MAH)

## Top Credible Scenarios

| Scenario | Description |
|---|---|
| Fire | Warehouse or reefer-stack fire; electrical cabinet fires; fuel fires |
| Hazardous Spills | Fuels, oils, chemicals, battery acid |
| Refrigerant Releases | A1 asphyxiants; A2L mildly flammable |
| Medical Emergencies | Serious injury/medical emergencies; heat illness |
| Security Events | Intrusion, theft, bomb/suspicious package, civil disturbance |
| Severe Weather | Lightning, high winds, flooding, storm surge |
| Water Incidents | Person overboard at quay |
| Equipment Failures | Crane/rigging failure; dropped container; stack collapse; plant rollover |
| Confined Space | Exposure/rescue (sumps, voids, reefer machinery spaces) |
| Cyber Incidents | Impacting safety systems, reefer monitoring, access control |
| Utility Loss | Black start; emergency power changeover |

## Seasonal Considerations

| Season | Risk | Mitigation |
|---|---|---|
| Rainy Season | Flooding risk | Pre-season stormwater inspections |
| Hot Season | Heatwaves | Work/rest cycles and hydration protocols |

# 6. Governance, Command, and Responsibilities

## 6.1 Command and Control

| Element | Description |
|---|---|
| ICS Structure | IC; Sections for Operations, Planning, Logistics, Finance/Admin; Safety Officer; PIO; NPA Liaison Officer. Scalable per incident |
| Command Posts | Primary CP: Terminal Control Room, Admin Block Level 1; Secondary CP: Security Gatehouse; Media Area: Visitor car park |
| Staging Areas | Yard staging bay north of Warehouse 1 |

## 6.2 Roles and Authorities

| Role | Responsibilities |
|---|---|
| Managing Director | Provides resources; approves ERP revisions; chairs GK & A side of Joint Emergency Steering Committee; authorizes escalations and CAPEX for life-safety |
| NPA Safety Director | Co-chairs governance; endorses revisions; ensures NPA alignment |
| Incident Commander | Declares emergencies; sets priorities; approves re-entry; orders evacuations and search/rescue; authorizes hot work post-incident via permit |
| Deputy IC | Ensures continuity of command; assumes IC role if needed |
| NPA Liaison Officer | Coordinates NPA Port Control, Port Police, Marine units; ensures statutory notifications to agencies |
| ERT Leaders | Task, supervise, and account for teams; ensure PPE and gas monitoring are used; report status to IC |
| Warehouse Controller | Executes emergency shutdown; secures hazardous storage; supports accountability |
| Reefer Maintenance Lead | Manages reefer power isolation/restore; engages certified refrigeration contractors; safeguards temperature-critical cargo |
| Security Supervisor | Establishes perimeters; manages access control and gate protocols; preserves evidence; liaises with NPA Port Police |
| PIO | Controls public/media/customer messaging; issues holding statements; maintains rumor control; ensures social media policy enforcement |
| All Personnel/Contractors | Raise alarms; stop work; evacuate; follow IC/ERT instructions; participate in drills; support headcounts. Non-compliance results in removal and disciplinary action |

## 6.3 Minimum Staffing and Duty Rosters

| Requirement | Details |
|---|---|
| On-shift ERT | Fire 3; HAZMAT 2; Medical 2; Security 2; Technical 1; IC/Deputy IC accessible within 10 minutes |
| Competency Currency | All on-shift ERT members shall be current on SCBA and gas monitoring within 12 months; first aiders with AED competency within 24 months |

## 6.4 Business Continuity Linkage

| Aspect | Description |
|---|---|
| BCP Tie-in | Reefer continuity, alternate cold storage, mutual aid for reefer power, client comms during prolonged outages |

# 7. Activation, Notification, and Initial Actions

## 7.1 Triggers

| Trigger | Description |
|---|---|
| Fire/Explosion | Fire/explosion/smoke alarm |
| Chemical Spill | Chemical or fuel spill |
| Refrigerant Leak | Refrigerant alarm/odor |
| Medical Emergency | Serious injury |
| Security Incident | Security incident |
| Severe Weather | Severe weather/flooding/lightning |
| Structural Threat | Structural compromise |
| Water Incident | Person overboard |
| Power Loss | Extensive power loss affecting reefers |

## 7.2 First Five Minutes (Mandatory)

| Action | Description |
|---|---|
| Raise Alarm | Pull station or radio in clear text "EMERGENCY – Location – Type" |
| Declare Emergency | IC or senior-most present declares emergency; stands up Command Post; assigns scribe; opens Incident Log with time-stamps |
| Dispatch ERT | Confirm PPE (SCBA for unknown atmospheres) |
| Notify Authorities | Notify NPA Port Control immediately; escalate to Fire/EMS/Police/LASEMA/LASWA/NIWA as needed per matrix |
| Secure Site | Lock down or open gates as appropriate to incident type to secure site or facilitate emergency ingress |

## 7.3 First Ten Minutes

| Action | Description |
|---|---|
| Evacuation/Shelter | Order evacuation or shelter-in-place as conditions dictate |
| Exclusion Zones | Establish exclusion zones; deny entry |
| Headcount | Achieve full headcount reconciliation at muster points within 10 minutes; identify missing persons; do not start rescue unless risk is ALARP and approved by IC |
| SITREP | Issue SITREP 1 to leadership and NPA |

### 7.4 Alerting Systems

| System | Description |
|---|---|
| Primary | Site sirens/PA; UHF radios (Channels: 1 Ops, 2 ERT, 3 Command); SMS/app alerts |
| Backup | Air horns, runners |

## 8. Emergency Response Procedures

### General Priorities

| Priority | Description |
|---|---|
| 1. Protect Life | Immediate protection of personnel |
| 2. Isolate Hazards | Contain and control the incident |
| 3. Prevent Escalation | Stop the incident from worsening |
| 4. Protect Environment | Prevent environmental damage |
| 5. Preserve Evidence | Maintain evidence for investigation |

### 8.1 Fire

| Aspect | Procedure |
|---|---|
| PPE | Full structural gear and SCBA for interior/close approach |
| Electrical Fires | Isolate power first if safe; use CO2/DCP; avoid water on live components |
| Flammable Liquid Fires | Apply foam; protect drains; deploy booms to capture runoff; avoid fluorinated foams unless no alternative; prefer fluorine-free foam (F3) |
| Warehouse/Stack Fires | Establish 360° size-up; protect exposures; set collapse zones; use TIC to confirm knockdown; ventilate cautiously to prevent flashover; compartment doors stay closed until cooled and monitored |
| Post-fire | Fire watch 60 minutes minimum; water/runoff containment and sampling; document all actions and foam use |

### 8.2 Hazardous Spills/HAZMAT

| Step | Action |
|---|---|
| Approach | Approach upwind/upstream; identify via SDS/IMDG; don chemical-resistant PPE; consider SCBA for unknown/toxic atmospheres |
| Control | Stop the source; isolate ignition sources; ground/bond for flammables; deploy drain covers, booms, absorbents, temporary bunds; protect watercourses |
| Collection | Overpack drums, labeled containers; segregate waste; store in designated HAZMAT area for licensed disposal |
| Notifications | NPA Port Control; LASEPA/NESREA where thresholds met; record volumes and materials |
| Decontamination | Establish gross decon for responders/casualties when required before EMS handover |

## 8.3 Refrigerant Leaks (Reefer Stacks)

### Hazards

| Hazard | Description |
|---|---|
| A1 | Oxygen displacement |
| A2L | Mild flammability; ignition risk; decomposition products if heated |

### Immediate Controls

| Control | Action |
|---|---|
| Evacuation | Evacuate immediate area; mark no-ignition zone |
| Cordon | Minimum cordon: 50 m if concentration and wind unknown; adjust based on readings |
| Hot Work | Prohibit hot work; isolate electrical power to affected unit(s) |

### Monitoring

| Equipment | Procedure |
|---|---|
| Leak Detector | Use calibrated refrigerant leak detector (NDIR/IR for halocarbons) plus multi-gas meter |
| Alarm Set-points | Warn at 10% LFL for A2L; evacuate/cordon at ≥20–25% LFL; manage A1 as asphyxiant—use SDS OEL (often ~1,000 ppm TWA; confirm per SDS) |

### Response

| Action | Description |
|---|---|
| Ventilation | Ventilate naturally or mechanically; ensure continuous ventilation in reefer alleys; interlock hot work shutdown where fixed detection exists |
| Repair | Only qualified technicians to isolate/repair; recover refrigerant to approved cylinders; record loss and service details |

### Re-occupy Only When

| Parameter | Acceptable Range |
|---|---|
| O2 | 19.5–23.5% |
| LEL | < 10% |
| Refrigerant | Below OEL per SDS |
| Stability | Two consecutive stable readings 5–10 minutes apart |

## 8.4 Medical Emergencies

| Step | Action |
|---|---|
| Scene Safety | Ensure scene safety; announce "Medical Emergency – General Hospital Ikorodu"; deploy AED/trauma kit |
| First Aid | DRSABC; control bleeding; burns: cool with clean water (≥20 minutes); chemical exposures: flush ≥15 minutes and consult SDS |
| Heat Stress | Remove from heat, active cooling, hydration, monitor vitals |
| EMS Coordination | Coordinate LASAMBUS/112/727; guide ambulance through designated gate; ensure casualty/witness logs; decon equipment after use |

## 8.5 Security Incidents

| Incident Type | Response |
|---|---|
| Lockdown | Lockdown/access control per PFSP; badge deactivation if required; ensure CCTV preservation and evidence handling |
| Bomb/Suspicious Package | Follow checklist; evacuate upwind; avoid radios within 15 m of suspect device |
| Intrusion/Theft | Do not pursue; preserve scene; notify NPA Port Police |
| Civil Disturbance | Shelter-in-place; safe room protocols; code words for silent alerts; comply with law enforcement |
| Communications | PIO manages all external comms; staff prohibited from social media posting |

## 8.6 Severe Weather and Flooding

### Monitoring

| Parameter | Action |
|---|---|
| Alerts | Lightning alerts; wind thresholds; tidal/surge data |

### Actions

| Action | Description |
|---|---|
| Secure Assets | Secure loose cargo/equipment; suspend high-risk work; cease crane operations above defined wind speeds |
| Protect Goods | Elevate vulnerable goods; protect critical records; shut non-essential power |
| Stormwater | Activate stormwater controls; deploy sandbags/booms; isolate drains (where valved) |
| Reefer Continuity | Transfer to emergency power per prioritization matrix; manual temp checks/logs during outages |
| Post-Event | Structural/electrical/environmental inspections; document findings and corrective actions |

## 8.7 Person Overboard

| Step | Action |
|---|---|
| Alert | Shout "Man Overboard"; maintain constant visual; point at casualty |
| Equipment | Throw lifebuoy with light/line; deploy additional flotation and throw-bags |
| Vessels | Halt nearby craft unless directed; notify NPA Marine/Police; launch designated rescue craft if safe; approach leeward; avoid crush points |
| Recovery | Treat for hypothermia/aspiration; call ambulance; preserve evidence; debrief |

## 8.8 Confined Space Incident/Rescue

| Requirement | Procedure |
|---|---|
| Entry Protocol | Do not enter without permit and rescue plan; perform atmospheric testing; continuous monitoring |
| Rescue | Rescue only by trained team with retrieval systems and standby; use SCBA or supplied air as required; keep entry/exit logs |

## 8.9 Crane/Load Handling Incidents

| Action | Description |
|---|---|
| Isolate | Establish collapse zone; isolate area; confirm structural stability |
| Recovery | Do not move load until assessed; engage specialist recovery; investigate rigging failure; preserve lifting gear as evidence |

## 8.10 Cyber Incident Impacting Safety/Operations

| Response | Action |
|---|---|
| Incident Response | Activate incident response; isolate affected networks; move to manual controls for safety-critical systems (reefer monitoring, access control) per BCP; maintain paper logs; notify leadership and NPA if port operations affected |

## 8.11 Utility Loss/Black Start

| Step | Action |
|---|---|
| Checklist | Black start checklist: verify generator readiness; initiate ATS/manual transfer; confirm priority loads (reefer racks, control room, lighting, comms); load shed per matrix; verify fuel autonomy and quality (water-in-fuel test); document times and loads |

# 9. Evacuation, Shelter-in-Place, Accountability, and Re-entry

## 9.1 Evacuation

### Triggers

| Trigger | Description |
|---|---|
| Fire | Fire incident |
| HAZMAT | Hazardous material/refrigerant release |
| Structural | Structural threat |
| Water | Person overboard in proximity |
| Security | As per IC directive |
| Weather | Severe weather conditions |

### Routes and Muster Points

| Element | Description |
|---|---|
| Routes | Use primary/secondary routes to four muster points located upwind/up-current and away from reefer banks and HAZMAT stores; maps in Appendix B; muster capacity limits posted |
| Assisted Evacuation | Buddy system; evacuation chairs where needed; Wardens/Marshals sweep zones, stairwells, amenities |
| Vehicles | Do not use unless directed for controlled egress or to clear critical paths |

## 9.2 Shelter-in-Place

### Trigger

| Trigger | Description |
|---|---|
| External Threat | External toxic plume; civil disturbance/armed threat; lightning with no safe egress |

### Actions

| Action | Description |
|---|---|
| Move Indoors | Move indoors; close doors/windows; switch off ventilation; seal gaps; stay tuned to Channel 3; await all-clear |

## 9.3 Accountability

| Procedure | Description |
|---|---|
| Headcount | Headcount completed within 10 minutes using T-cards, access logs, visitor registers, PTW lists; report missing persons with last known location; IC approves search-and-rescue only when risk is ALARP |

## 9.4 Re-entry and Re-energization

### IC grants re-entry with NPA concurrence after

| Requirement | Standard |
|---|---|
| Atmosphere | O2 19.5–23.5%; LEL < 10%; CO < 25 ppm; H2S/other toxics below OEL per SDS; refrigerant below OEL per SDS; two consecutive stable readings 5–10 minutes apart |
| Physical | No visible smoke/heat; structural integrity confirmed; utilities safe; spills neutralized/contained; drains protected |

### Phased Re-entry

| Phase | Description |
|---|---|
| Essential | Essential personnel first |
| Full | Full re-energization via permit-to-work and hot work permits as applicable |

### Post Re-entry

| Action | Description |
|---|---|
| Fire Watch | 60-minute fire watch and status update to all staff/clients as needed |

# 10. Communications and Information Management

## 10.1 Radio and Channels

| Channel | Use |
|---|---|
| 1 | Operations |
| 2 | ERT |
| 3 | Command |
| Protocol | Plain language; call signs; minimize non-essential traffic. Radio fallback: spare radios, charged batteries, cellular PTT if available |

## 10.2 Notifications and Scripts

| Action | Description |
|---|---|
| Authorities | IC or delegate notifies NPA Port Control; Lagos State Fire and Rescue; LASEMA/NEMA; LASAMBUS (112/727); Nigeria Police (including Marine); LASEPA/NESREA; LASWA/NIWA; power utility; mutual aid |
| Scripts | Use standard PA/holding statements (Appendix F) |

## 10.3 Public Information

| Protocol | Description |
|---|---|
| PIO Only | Only PIO communicates externally; all statements approved; social media posting by staff prohibited |

## 10.4 Information Governance

| Element | Description |
|---|---|
| Records | Incident Log and Situation Reports (SITREPs) maintained; evidence preserved (CCTV, photos, samples) with chain-of-custody |
| Data Protection | Handle personal data and CCTV per NDPA; restrict access; define retention and deletion for non-incident footage |

## 10.5 Records Retention

| Requirement | Duration |
|---|---|
| Incident Records | Minimum five years for incident records, training, detector calibrations, environmental manifests |
| Extensions | Longer if required by regulator/insurer |

# 11. Resources, Equipment, and Maintenance

## 11.1 Minimum Critical Inventory

### Firefighting

| Equipment | Minimum |
|---|---|
| DCP/CO2/foam extinguishers | By hazard class |
| Hydrants/hose reels | As required |
| Fire pumps | As required |
| Foam trailer | F3 preferred |
| Thermal imaging cameras | As required |

### HAZMAT

| Equipment | Minimum |
|---|---|
| Spill kits | Oil/chemical/universal |
| Drain covers | As required |
| Absorbent booms | As required |
| Overpack drums | As required |
| Neutralizers | As required |

## Detection

| Equipment | Minimum |
|---|---|
| Multi-gas meters | O2, LEL, CO, H2S |
| Refrigerant detectors | Compatible with installed gases |
| Fixed detection | Where installed |

## Respiratory/Rescue

| Equipment | Minimum |
|---|---|
| SCBA sets | With spare cylinders |
| Escape hoods | As required |
| Lifebuoys | With lights/lines every 25–50 m |
| Throw-bags | As required |
| Quay ladders | As required |
| Rescue boat | Readiness |

## Medical

| Equipment | Minimum |
|---|---|
| First-aid kits | As required |
| Trauma kits | As required |
| AEDs | As required |
| Stretchers/spine boards | As required |
| Eyewash/shower stations | As required |

## Power

| Equipment | Minimum |
|---|---|
| Emergency generators | With ATS prioritizing reefer circuits, control room, lighting, comms |
| Fuel autonomy | ≥ 48 hours |
| Critical spares | ATS controller, breakers, reefer plugs/connectors |

### 11.2 Maintenance and Assurance

| Activity | Frequency |
|---|---|
| CMMS inspections | As scheduled |
| Visual checks | Monthly |
| Functional tests | Quarterly |
| Detector bump tests | Per shift use |
| Calibration | Per manufacturer |
| SCBA flow tests | As required |
| AED maintenance | Pads/battery within expiry |
| Hydrant tests | Flow tests |
| Generator tests | Monthly load tests; quarterly full-load bank tests |
| Color-tag system | In service/out of service |
| Third-party service | Contracts for pumps, detectors, SCBA, AEDs, generators, reefer systems |

## 12. Training, Competency, and Drills

### 12.1 Induction (Mandatory for All Before Access)

| Topic | Description |
|---|---|
| Site Hazards | Overview of site hazards |
| Alarms | Emergency alarm procedures |
| Muster Points | Location and use of muster points |
| Stop-work | Authority to stop unsafe work |
| Incident Reporting | How to report incidents |
| PPE | Personal protective equipment requirements |
| PTW | Permit-to-work system |
| Social Media | Restrictions on social media use |

## 12.2 Core Training (All Staff)

| Training | Description |
|---|---|
| Fire Extinguisher | Use of portable fire extinguishers |
| Spill Awareness | Recognition and basic response to spills |
| First-Aid Awareness | Basic first aid awareness |
| Security Awareness | Security awareness training |
| Emergency Comms | Emergency communication procedures |
| Shelter-in-Place | Shelter-in-place procedures |

## 12.3 Role-Specific Competencies

### ERT

| Competency | Description |
|---|---|
| Advanced Firefighting | Industrial/port firefighting techniques |
| HAZMAT Spill Control | Spill containment and control |
| SCBA | Self-contained breathing apparatus use |
| Confined Space | Confined space awareness |
| Water Rescue | Water rescue elements |
| Incident Command | ICS 100/200; ICS 300 for IC/Deputy |
| Refrigerant Hazards | Refrigerated cargo hazards |

### Operators

| Competency | Description |
|---|---|
| Equipment | Forklift/crane operation |
| Banksman | Banksman/signaller training |
| Load Securement | Cargo load securement |
| Work at Height | Work at height safety |
| LOTO | Lockout/tagout procedures |

### Technicians

| Competency | Description |
|---|---|
| Refrigerant Safety | Refrigerant safety including A2L awareness |
| Electrical Safety | Electrical safety procedures |
| Arc-flash | Arc-flash awareness |
| Hot Work | Hot work permit system |
| LOTO | Lockout/tagout procedures |

### Wardens/Marshals

| Competency | Description |
|---|---|
| Evacuation Leadership | Leading evacuations |
| Headcount | Headcount procedures |
| Crowd Management | Crowd management techniques |
| Assisted Evacuation | Assisting persons with mobility issues |

### PIO/Comms

| Competency | Description |
|---|---|
| Crisis Comms | Crisis communication techniques |
| Media Handling | Media handling procedures |
| Message Maps | Creating and using message maps |
| Rumor Control | Controlling rumors during emergencies |

## 12.4 Drill Program and Cadence

| Drill Type | Frequency |
|---|---|
| Site-wide Evacuation | Quarterly (include 1 night-shift and 1 no-notice annually) |
| HAZMAT/Refrigerant | Semi-annual (at least one A2L scenario) |
| Multi-agency Full-scale | Annual with NPA and emergency services (include water rescue) |
| Table-top | Quarterly for leadership/supervisors with injects |

## 12.5 KPIs and Records

| KPI | Target |
|---|---|
| Headcount Time | Within 10 minutes |
| Response Times | As defined |
| Participation | ≥ 90% on-duty participation |
| Corrective Actions | Closed ≤ 30 days |
| Requalification | Before expiry |
| Records | Training matrix and certificates maintained |

# 13. Post-Incident Actions and Reporting

## 13.1 Stabilization and Handover

| Action | Description |
|---|---|
| Scene Safety | Confirm scene safety |
| Command Retention | Maintain command until formal transfer to authorities |
| Documentation | Document handover times and accepting authority |

## 13.2 Notifications

| Recipient | Timeline |
|---|---|
| NPA | Immediate |
| Environmental | LASEPA/NESREA within prescribed periods |
| Insurers | Within policy terms |
| Clients | Per contract/PIO guidance |

## 13.3 Investigation

| Element | Description |
|---|---|
| Lead Investigator | Appointed by HSE Manager |
| Methods | 5 Whys, bow-tie, fault tree |
| Evidence | Preserve evidence with chain-of-custody |
| Interviews | Documented interviews |
| Timeline | Reconstructed via CCTV/access logs |

## 13.4 Corrective and Preventive Actions (CAPA)

| Action | Description |
|---|---|
| Assignment | Assign owners and due dates |
| Verification | Verify effectiveness |
| Management Review | Include in management review |
| ERP Updates | Update ERP/SOPs within 30 days where required |
| MOC | Raise MOC for systemic changes |

## 13.5 Environmental Remediation

| Activity | Description |
|---|---|
| Sampling | Sampling/monitoring |
| Regulator | Regulator engagement |
| Contractors | Licensed waste contractors with manifests retained |

## 13.6 Learning and Reporting

| Output | Description |
|---|---|
| Final Report | Final incident report |
| Regulatory | Regulatory submissions |
| Insurer | Insurer loss pack |
| Client | Client reports |
| Lessons | Lessons shared via safety alerts/toolbox talks |

# 14. Financial Provisioning and Insurance

| Element | Description |
|---|---|
| Emergency Funding | Pre-approved thresholds for rapid procurement and contractor call-outs; emergency vendor list activated by IC/Procurement |
| Property Damage | Property damage/BI insurance |
| Machinery Breakdown | Machinery breakdown (generators/reefer electrical) |
| Terminal Liability | Terminal operators liability/bailee's liability |
| Employers' Liability | Employers' liability/workers' compensation |
| Public Liability | Public liability |
| Environmental | Environmental impairment/pollution liability |
| Marine Cargo | Marine cargo as applicable |
| Claims Management | Immediate loss notice; mitigation; evidence preservation; adjuster coordination; cost tracking; salvage/repair decisions; claims coordinator designated |

## 15. Plan Governance, Review, and Distribution

| Element | Description |
|---|---|
| Ownership | Plan Owner HSE Manager; approval by Managing Director; NPA Safety concurrence recorded |
| Review Cycle | Annual or after major incident, regulatory change, significant exercise findings, or material facility/process change |
| Document Control | Versioning; revision log; change summaries and approvers; controlled copies in Control Room, Security Gatehouse, ERT lockers; digital repository access-controlled |
| Assurance | Internal audits and joint NPA reviews; KPI dashboard to leadership; actions tracked to closure |
| MOC | Required for any change affecting layout, risks, alarms, egress, utilities, or emergency power priorities; training and communication delivered before change goes live |

## 16. Appendices

| Appendix | Content |
|---|---|
| A | Contact and Notification Matrix (internal, regulators, emergency services, mutual aid, utilities, insurers; time limits and scripts) |
| B | Site Plans, Zoning, Evacuation Routes, Muster Points; hydrants, shut-offs, reefer plug racks; Command Post and Staging; Media Area |
| C | Emergency Resources Inventory and Maintenance Schedule (minimum quantities; inspection/calibration/test cadences; tagging rules) |
| D | Training Matrix and Role Competency Profiles (currency/recert intervals; duty roster coverage) |
| E | Checklists and Forms (activation; size-up; headcount; search-and-rescue authorization; re-entry; hot work and gas test; SITREP; investigation; CAPA; decon; bomb threat) |
| F | Standard PA Scripts, Alarm Tones, and Media Holding Statements (fire, spill, medical, severe weather, security lockdown, reefer service interruption) |
| G | Chemical/Refrigerant Inventory and SDS Index (OELs, LFLs, detector compatibility) |
| H | Regulatory/Standards Reference List and Compliance Mapping (NPA, ISPS, IMDG, LASEPA/NESREA, LSSC, LASWA/NIWA, EN 378, NFPA) |
| I | Insurance Certificates and Broker/Insurer Contacts (notification windows, claim pack contents) |
| J | Mutual-Aid Agreements/MoUs and Service SLAs (refrigeration contractors, spill response, fuel supply, heavy lift/crane recovery) |
| K | Glossary and Acronyms |
| L | Records Retention Schedule (including NDPA data governance) |

## Compliance Statement

Compliance with this Plan is compulsory and jointly enforced by GK & A and NPA. All personnel, contractors, and visitors are bound by these requirements. Any deviation, delay, or shortcut is a breach of GK & A and NPA mandates and will trigger immediate corrective measures and disciplinary action, up to removal from site and contract termination for cause. Continuous readiness, rapid notification, disciplined execution, and complete documentation are non-negotiable.

## 90-Day Implementation Roadmap

| Task | Timeline |
| --- | --- |
| Populate Appendices | A–G with site-specific names, 24/7 numbers, maps, inventories, and forms |
| Confirm Locations | Command Post, Staging, and Media Areas on plans; post evacuation maps in English and Yoruba; show prevailing wind/tidal directions |
| Alert Systems | Verify siren/PA coverage; prepare backup alert kits (air horns, runners) |
| Equipment Inventory | Inventory and close gaps on critical equipment (SCBA, detectors, spill kits, foam F3, life rings/throw-bags, AEDs, reefer plugs/spares) |
| Calibration | Calibrate/bump-test detectors; label due dates; stock calibration gases |
| Duty Rosters | Finalize duty rosters; minimum on-shift ERT coverage; publish escalation tree |
| Power Systems | Test ATS, black start, and 48-hour fuel autonomy; define load-shedding and reefer prioritization matrix |
| Tabletop Exercise | Execute a tabletop within 30 days (scenario: refrigerant leak + concurrent medical) |
| Evacuation Drill | Conduct a night-shift evacuation drill within 60 days |
| Full-scale Drill | Schedule a full-scale multi-agency drill within 90 days (include water rescue and reefer incident) |
| Communications | Issue social media restrictions reminder; brief PIO and Client Relations on scripts |
| CAPA Tracker | Launch CAPA tracker; report progress monthly to the Joint Emergency Steering Committee |

# Appendix A — Contact and Notification Matrix (External Authorities and Insurers)

## A.1 External Authorities (Ikorodu/Lagos)

| Authority | Contact | Details |
|---|---|---|
| NPA Port Control (Ikorodu) | Phone: +234 901 222 3101 | Email: ikorodu.portcontrol@nigerianports.gov.ng (verify with NPA Liaison) |
| NPA Port Police (Ikorodu) | Phone: +234 902 222 3102 | Email: portpolice.ikorodu@nigerianports.gov.ng (verify with NPA Liaison) |
| Nigeria Police Marine/Water Unit | Phone: +234 903 222 3103 | Email: info@npf.gov.ng (use phone for activation; email for follow-up) |
| Lagos State Fire and Rescue Service | Phone: +234 904 222 3104 | Email: info@lsfrs.lagosstate.gov.ng (verify current address; activation via 112/727 only) |
| LASAMBUS/Ambulance (Lagos) | Phone: +234 905 222 3105 | Email: info@health.lagosstate.gov.ng (use 112/727 for activation; email for admin follow-up) |
| LASEMA (Emergency Management Agency) | Phone: +234 906 222 3106 | Email: info@lasema.lagosstate.gov.ng |
| NEMA (Federal) | Phone: +234 907 222 3107 | Email: info@nema.gov.ng |
| LASEPA (Environmental Protection Agency) | Phone: +234 908 222 3108 | Email: info@lasepa.lagosstate.gov.ng |
| NESREA (Federal Environmental Regulator) | Phone: +234 909 222 3109 | Email: info@nesrea.gov.ng |
| LASWA (Waterways Authority) | Phone: +234 910 222 3110 | Email: info@laswa.lagosstate.gov.ng |
| NIWA (National Inland Waterways Authority) | Phone: +234 911 222 3111 | Email: info@niwa.gov.ng |
| Power utility control room (Ikorodu area – likely Ikeja Electric) | Phone: +234 912 222 3112 | Email: customercare@ikejaelectric.com (verify service coverage for your feeders) |
| Mutual Aid — Ikorodu Jetty Operators Association | Phone: +234 913 222 3113 | Email: admin@ijoa.org.ng (verify) |
| Marine Tug/Rescue — Ikorodu Marine Services | Phone: +234 914 222 3114 | Email: ops@ikorodumarine.com.ng (verify) |
| Industrial Fire Support (Regional) | Phone: +234 915 222 3115 | Email: duty@regionalfiresupport.com.ng (verify) |

## A.2 Insurance and Claims

| Insurer | Contact | Details |
| --- | --- | --- |
| Broker — NaijaRisk Brokers (Primary) | Phone: +234 701 444 5101 | Email: service@naijariskbrokers.com.ng (verify) |
| Broker — NaijaRisk Brokers (Backup) | Phone: +234 702 444 5102 | Email: claims@naijariskbrokers.com.ng (verify) |
| Property/Business Interruption — Guardian Assurance | Phone: +234 703 444 5103 | Policy: TBD-PA-2025-001; Email: claims@guardianassurance.com.ng (verify) |
| Machinery Breakdown — Atlas Engineering Ins. | Phone: +234 704 444 5104 | Policy: TBD-MB-2025-002; Email: claims@atlasengineeringins.com.ng (verify) |
| Terminal Operators/Bailee's Liability — Coastal Liability Ins. | Phone: +234 705 444 5105 | Policy: TBD-TL-2025-003; Email: claims@coastalliability.com.ng (verify) |
| Employers' Liability/Workers' Comp — Safeguard Ins. | Phone: +234 706 444 5106 | Policy: TBD-EL-2025-004; Email: claims@safeguardins.com.ng (verify) |
| Public Liability — Guardian Assurance | Phone: +234 707 444 5107 | Policy: TBD-PL-2025-005; Email: liability@guardianassurance.com.ng (verify) |
| Environmental Impairment/Pollution — GreenShield Ins. | Phone: +234 708 444 5108 | Policy: TBD-EP-2025-006; Email: envclaims@greenshieldins.com.ng (verify) |
| Marine Cargo (if applicable) — OceanCover | Phone: +234 709 444 5109 | Policy: TBD-MC-2025-007; Email: claims@oceancover.com.ng (verify) |

# Appendix B — Site Plans, Zoning, Evacuation, and Key Locations

## B.1 Site Identification

| Element | Details |
|---|---|
| Facility | GK & A Terminal, NPA Lighter Terminal (Ikorodu) |
| Address | NPA Lighter Terminal Complex, Ipakodo Road, Ipakodo, Ikorodu, Lagos State, 104101, Nigeria |
| Main Gate GPS | 6.5990068, 3.4814838 |
| Emergency Activation | Dial 112 or 727 |
| Site Emergency Email | info@gkaport.com |

## B.2 Command, Staging, and Media Points

| Location | Details |
|---|---|
| Primary Command Post (CP) | Terminal Control Room, Admin Block Level 1; Reference: Admin Block near Main Gate at 6.5990068, 3.4814838; Features: Hard-wired power/UPS; dedicated radios/spares; wall site plan; Incident Log kit; gas monitor cache |
| Secondary CP | Security Gatehouse (co-located at Main Gate); Reference: 6.5990068, 3.4814838; Features: Redundant radios; CCTV access; duplicate muster board; basic logs |
| Staging Area S1 | Yard staging bay north of Warehouse 1; Reference: Signed "S1—Staging"; cones and signage stored at Gatehouse |
| Media Area | Visitor car park outside the operations fence near Main Gate; Reference: Signed "Media Area—PIO Control Only"; segregated from muster points |

## B.3 Evacuation Routes and Muster Points

| Element | Description |
|---|---|
| Primary Routes | From warehouses and reefer alleys to North Car Park and West Laydown via signed paths; avoid reefer stacks and HAZMAT stores |
| Secondary Routes | Southern and eastern egress to South Green Space and East Open Ground |
| Muster Points | A: North Car Park (capacity 120); B: West Laydown Yard (capacity 120); C: South Green Space (capacity 150); D: East Open Ground (capacity 150) |
| Accessibility | Evacuation chairs at Admin Block stairwell and Warehouse 1 mezzanine; wardens sweep amenities/stairwells before leaving |
| Signage | Bilingual English/Yoruba; reflective; "Avoid Reefer Stacks" icons |

## B.4 Fire Water Supplies and Critical Utilities

| Equipment | Location/Details |
|---|---|
| Hydrants/Monitors | H1 Quay NW; H2 Quay mid-berth; H3 Quay SE; H4 Perimeter west curve; H5 W1 north wall; H6 W1 south wall; H7 Admin frontage; H8 Fuel point bund (outside); H9 Reefer aisle; H10 Yard east perimeter; Fixed monitors: M1 Quay NW; M2 Quay SE |
| Fire Pump Room(s) | PR-1: Annex south of Warehouse 1; diesel duty + electric standby; access clear; keys at Gatehouse and Control Room |
| Electrical Main Isolation | ELV-1: Main LV room by Control Room; lockable main incomers; LOTO points; YDP-1: Yard distribution panel (west yard) feeding reefer racks and lighting; labeled feeders |
| Reefer Plug Racks | R1: Western reefer alley Row 1; R2: Eastern reefer alley Row 2; spares in Technician cage near ELV-1 |

## B.5 HAZMAT Storage and Environmental Controls

| Location | Details |
|---|---|
| Hazardous Stores | HS-1 Battery electrolyte/acid cabinet—Workshop annex (bunded, ventilated); HS-2 Paints/solvents flame-proof cabinet—W1 north alcove; HS-3 Oils/greases—Workshop lubricants rack |
| Fuel Points/Tanks | FP-1 Diesel tank + dispensing—West yard; bonded/grounded; spill berm intact |
| Storm Drains/Outfalls | Drains D1–D6 marked "Drain—Cover Here"; drain covers stored Gatehouse cage DC-1; Outfall OF-1: Perimeter culvert to waterway; deploy booms on risk |
| Spill Kit Stations | SK-1 Gatehouse; SK-2 W1 north door; SK-3 Workshop; SK-4 Reefer west alley; SK-5 Fuel point FP-1 |

## B.6 Quay and Water Safety

| Equipment | Location |
|---|---|
| Quay Ladders | L1 Quay NW; L2 Mid-berth; L3 Quay SE |
| Lifebuoys with Lights/Lines | LB-1 to LB-8 along quay; monthly inspection tags |
| Throw-bags/Rescue Poles | RB-1 Gatehouse; RB-2 Quay mid-berth kiosk; RB-3 W1 south door |
| Rescue Boat | Mid-berth cleat cluster; signed launch path; weekly outboard checks |

## B.7 Prevailing Conditions and Shelter-in-Place

| Condition | Details |
|---|---|
| Wind | Predominantly SW→NE; rainy-season squalls possible—IC selects upwind CP/muster |
| Tide/Current | Ebb/flood directions noted on daily ops board; watch flood-prone drains D4–D6 |
| Indoor Shelter Rooms | S-1 Admin Ground-Floor Meeting Room—limited windows; HVAC isolation switch; S-2 Warehouse 1 Office Core—solid walls; doors sealable; eyewash W1-EW-2 nearby |

## B.8 Traffic and Access Control

| Element | Details |
|---|---|
| Gatehouses/Access | Main Gate (6.5990068, 3.4814838) primary; East Gate normally closed—open only by Security Supervisor |
| Emergency Lanes/One-Way | Red-lined lanes along W1 north edge and quay spine; keep clear; removable bollards for Fire/EMS ingress |
| Crowd-Control Barriers | Stored Gatehouse Store GC-1; deploy to cordon Media Area/exclusion zones |

## B.9 Map Legend

| Symbol | Meaning |
|---|---|
| CP | Command Post (star) |
| Staging | Staging Area (triangle) |
| Media | Media Area (square) |
| Muster A–D | Muster Points (circles) |
| Hydrants/Monitors | Fire water supplies |
| Fire Pumps | Fire pump locations |
| Electrical Isolation | Electrical isolation points |
| Reefer Racks | Reefer plug rack locations |
| HAZMAT Stores | Hazardous material storage |
| Fuel Points | Fuel storage and dispensing |
| Drains/Outfalls | Storm drains and outfalls |
| Life Rings/Ladders | Water safety equipment |
| Shelter Rooms | Indoor shelter locations |
| Access Gates | Entry/exit points |
| Primary/Secondary Routes | Evacuation routes |

# Appendix C — Emergency Resources Inventory and Maintenance Schedule

## C.1 Firefighting Equipment

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| Portable Extinguishers | | | |
| DCP 9 kg | 40 | TBD | Monthly visual; Annual service; Hydrotest 5-year |
| CO2 5 kg | 25 | TBD | Monthly visual; Annual service; Hydrotest 5-year |
| Foam 9 L | 20 | TBD | Monthly visual; Annual service |
| Hose Reels | 20 | TBD | Monthly visual; Annual flow test |
| Hydrants | 18 | TBD | Quarterly flow test; Annual full-system test |
| Foam Concentrate | 1,000 L | TBD | Annual sample QA |
| Thermal Imaging Cameras | 2 | TBD | Quarterly function check |
| Color Tagging | Green—in service; Red—OOS with defect tag and work order ID | | |

## C.2 Spill/HAZMAT Equipment

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| Spill Kits | | | |
| Oil | 12 | TBD | Inspect monthly; Restock after use |
| Chemical | 8 | TBD | |
| Universal Absorbent Pads | 30 bales | TBD | |
| Booms | 450 m | TBD | 300 m oil-only; 150 m chemical |
| Drain Covers | 20 | TBD | |
| Overpack Drums | 12 | TBD | |
| Neutralizers | As needed | TBD | Battery acid, caustic, solvent; SDS stored with kits |
| Anti-Sparking Tools | 2 sets | TBD | |

## C.3 Detection and Monitoring

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| Multi-Gas Meters | 8 | TBD | Bump test before use/shift; Calibrate quarterly |
| Refrigerant Detectors | 4 portable | TBD | Calibrate per OEM |
| Fixed Detection | As installed | TBD | Quarterly function test; Annual calibration |
| Calibration Gases | As needed | TBD | Stock and log expiry dates |

## C.4 Respiratory/Rescue and Water Safety

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| SCBA Sets | 8 | TBD | Monthly checks; Annual flow test; Spare cylinders 16 |
| Escape Hoods | 20 | TBD | |
| Lifebuoys with Lights/Lines | Every 25–50 m | TBD | Monthly inspection |
| Throw-bags | 10 | TBD | |
| Quay Ladders | Every ~50 m | TBD | Quarterly inspection |

## C.5 Medical

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| AEDs | 4 | TBD | Pads/batteries within expiry; Monthly test |
| Trauma Kits | 8 | TBD | |
| Spine Boards/Stretchers | 4 | TBD | |
| Eyewash/Showers | 8 stations | TBD | Weekly activation; Annual service |

## C.6 Power Continuity

| Equipment | Minimum | Current | Maintenance |
|---|---|---|---|
| Generators | 2 x [kVA TBD] | TBD | Monthly start test; Quarterly load bank; Fuel autonomy ≥ 48 h; Water-in-fuel checks |
| ATS | As required | TBD | Functional test quarterly; Critical spares: controller and contactors |
| Reefer Plug Racks | As installed | TBD | Current plug count: TBD; Spare plugs/connectors 20 |

## C.7 Maintenance and CMMS Controls

| Activity | Description |
| --- | --- |
| Inspections | Entered into CMMS within 24 h; defects assigned with SLA |
| OOS Equipment | Escalated same day to HSE Manager and IC |
| Third-party Service | Certificates filed and auditable |

# Appendix D — Training Matrix and Competency Profiles

## D.1 Roles and Mandatory Competencies

| Role | Competencies | Refresh |
|---|---|---|
| Incident Commander/Deputy IC | ICS 100/200 (IC), ICS 300 (IC/Deputy); Emergency decision-making and dynamic risk assessment; Hot work authorization and gas testing | ICS: 3 years; Others: Annual |
| ERT Fire | Advanced firefighting (industrial/port); SCBA user; Thermal imaging, foam operations, exposure protection | Firefighting: 2 years; Others: Annual |
| ERT Spill/HAZMAT | HAZMAT awareness/operations; Spill containment/booming; drain protection; decon; SDS/IMDG interpretation | 2 years; Annual |
| ERT Medical | First Aid/CPR/AED; Trauma management/bleeding control; Heat stress recognition/management | 2 years; Annual |
| ERT Security | Access control/crowd management; Evidence preservation and PFSP interfaces; Bomb threat and suspicious package protocols | 2 years; Annual |
| Technical/Reefer | Electrical safety, LOTO; Refrigerant safety (A1/A2L), gas detection; Emergency power/ATS operations | Annual |
| Wardens/Marshals | Evacuation leadership/headcount; Assisted evacuation | Annual |
| Operators | Equipment certification; Banksman/slinging as applicable | 1–3 years; 2 years |
| All Staff and Contractors | Site induction and emergency orientation; Fire extinguisher practical; Spill awareness; Security awareness and social media restrictions | Before access/2 years; Annual |

## D.2 Drill Cadence (Minimum)

| Drill Type | Frequency |
|---|---|
| Evacuation | Quarterly (include 1 night-shift and 1 no-notice annually) |
| HAZMAT/refrigerant | Semi-annual (include A2L scenario if applicable) |
| Full-scale multi-agency | Annual with NPA Fire/Police, LASEMA, marine rescue |
| Table-top for leadership | Quarterly with injects |

## D.3 Duty Roster Coverage (Minimum)

| Requirement | Minimum |
|---|---|
| On-shift ERT | Fire 3; HAZMAT 2; Medical 2; Security 2; Technical 1; IC/Deputy reachable ≤ 10 minutes |
| First aiders per shift | ≥ 2 per zone (warehouse, yard/quay, admin) |

## D.4 Records and KPIs

| KPI | Target |
|---|---|
| Training currency | ≥ 95% |
| Drill participation | ≥ 90% of on-duty |
| Corrective actions closed | ≤ 30 days |

# Appendix G — Chemical and Refrigerant Inventory and SDS Index

## G.1 Inventory Fields

| Field | Description |
|---|---|
| Material | Chemical or refrigerant name |
| UN/IMDG/PG | UN number, IMDG class, Packing Group |
| CAS | Chemical Abstracts Service number |
| Storage location | Where material is stored |
| Max quantity | Maximum quantity allowed |
| SDS version/date | Safety Data Sheet version and date |
| OEL TWA/STEL | Occupational Exposure Limit - Time Weighted Average/Short Term Exposure Limit |
| LFL/UFL | Lower/Upper Flammability Limit |
| Vapor density | Vapor density relative to air |
| Water reactivity | Reactivity with water |
| Detector compatibility | Compatible detection equipment |
| Primary hazards/PPE | Primary hazards and required PPE |
| First-aid notes | First-aid procedures |
| Spill kit required | Required spill kit type |
| Segregation | Required segregation from other materials |
| Notification thresholds | Thresholds for regulatory notification |

## G.2 Example Entries (Replace with Actuals)

| Material | Details |
|---|---|
| Diesel Fuel | UN 1202; IMDG 3; PG III; CAS 68334-30-5; Storage: Fuel tank FT-1 (bunded), Yard West; Max qty: TBD L; SDS date: TBD; OEL: Oil mists 5 mg/m3 TWA; LFL/UFL: ~0.6–7.5% vol; Vapor density: >1; Detector: LEL (catalytic/IR); PPE: Anti-static PPE; nitrile gloves; goggles; Spill kit: Oil-only; booms; drain covers; foam for fire as needed; Segregation: Away from oxidizers |
| Lubricating Oil | Not regulated as DG (check SDS); Storage: Workshop rack; Max qty: TBD L; OEL: Oil mists 5 mg/m3; Spill kit: Oil-only |
| Battery Electrolyte | UN 2796; IMDG 8; PG II; CAS 7664-93-9; Storage: Battery room; acid cabinet; bunded; Max qty: TBD L; OEL: 0.2 ppm STEL (check SDS); Detector: Acid gas; PPE: Acid-resistant gloves/apron/face shield; Neutralizer: Sodium bicarbonate; copious water; Segregation: Away from bases/metals; Notification: LASEPA if > TBD L enters environment |
| Paints/Solvents | UN 1263; IMDG 3; PG II/III (varies); Storage: Paint locker (ventilated/flame-proof); LFL: ~1–1.4%; Controls: No ignition sources; bonding/grounding |
| Cleaning Chemicals | SDS-driven hazards; segregate from acids; eyewash nearby |
| Refrigerants | R-134a (A1, non-flammable); Asphyxiant at high conc.; OEL: ~1,000 ppm TWA (check SDS); Storage: in-use and spare cylinders in ventilated cage; Detector: IR; R-404A (A1); OEL: ~1,000 ppm TWA; R-452A (A1); OEL: per SDS; If A2L present (e.g., R-1234yf/ze): mildly flammable; monitor %LFL; ensure ventilation and no ignition sources |

## G.3 SDS Management

| Activity | Description |
|---|---|
| Master Index | Accessible to staff; SDS at point-of-use; QR codes optional; annual review |

## G.4 Segregation and Labeling

| Requirement | Description |
|---|---|
| IMDG Segregation | Per class; corrosives away from bases/flammables; oxidizers isolated |
| GHS Labeling | GHS labeling; secondary containers labeled |
| Storage Rooms | Ventilated with eyewash/shower nearby |

## G.5 Environmental Thresholds and Notifications

| Requirement | Description |
|---|---|
| Reportable Quantities | Define with LASEPA/NESREA (confirm with regulators) |
| Immediate Notification | Phone notification for any release to waterway |
| Written Reports | Follow statutory written report timelines |